

Robust Real-Time Facial Emotion Recognition

via Adversarial Feature Learning
and GAN-based Domain Adaptation

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The Great Data Drought of 2026

Regulatory Impact on Biometric Data

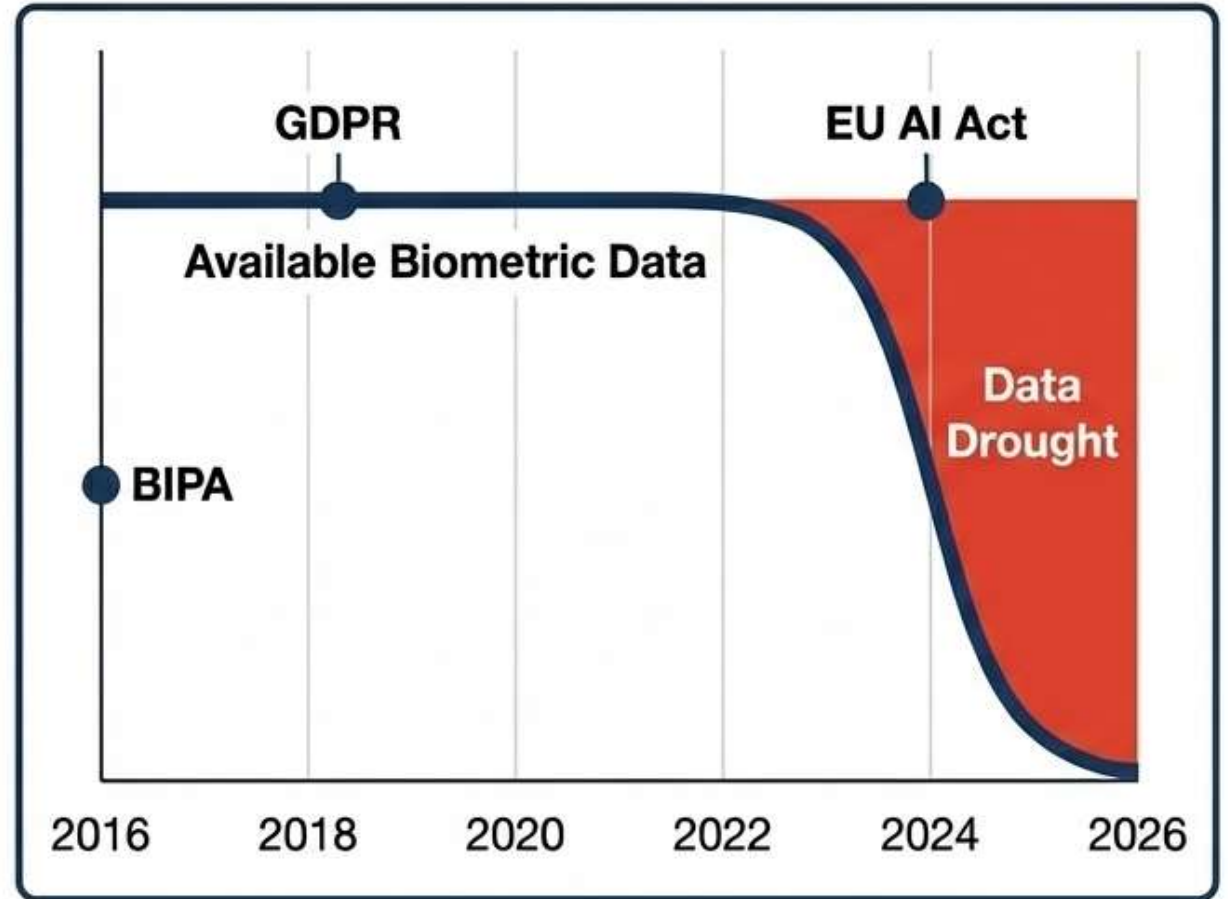
- The Industry Crisis:

Web-scraping for human faces is no longer legally viable.

- Regulatory Landscape:

GDPR, EU AI Act 2024, and BIPA have forced massive benchmarks offline.

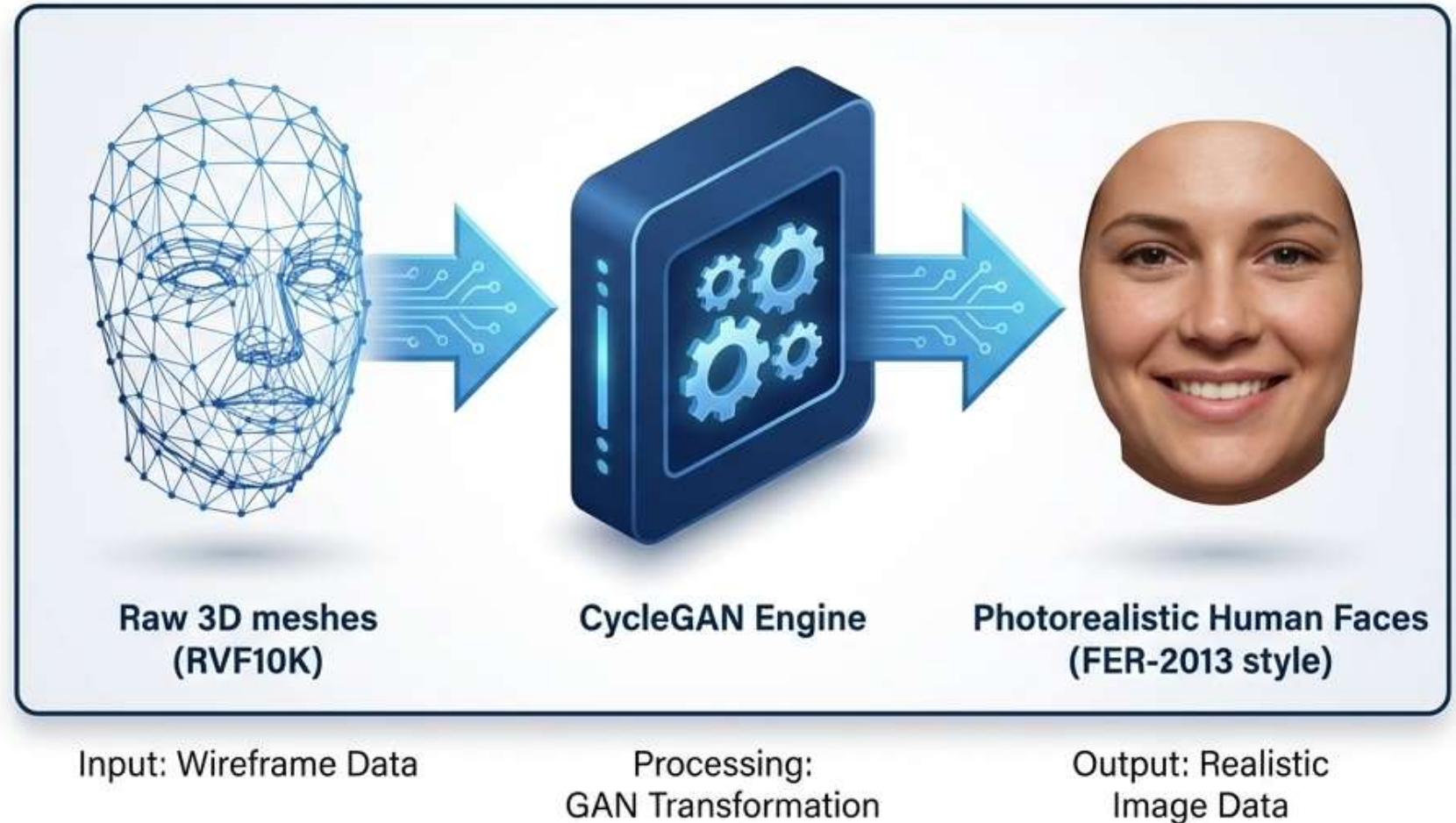
- **Result:** Critical “Data Drought” bottlenecking robust FER development.



The Synthetic Gold Rush

GANs as the "New Data Oil Well"

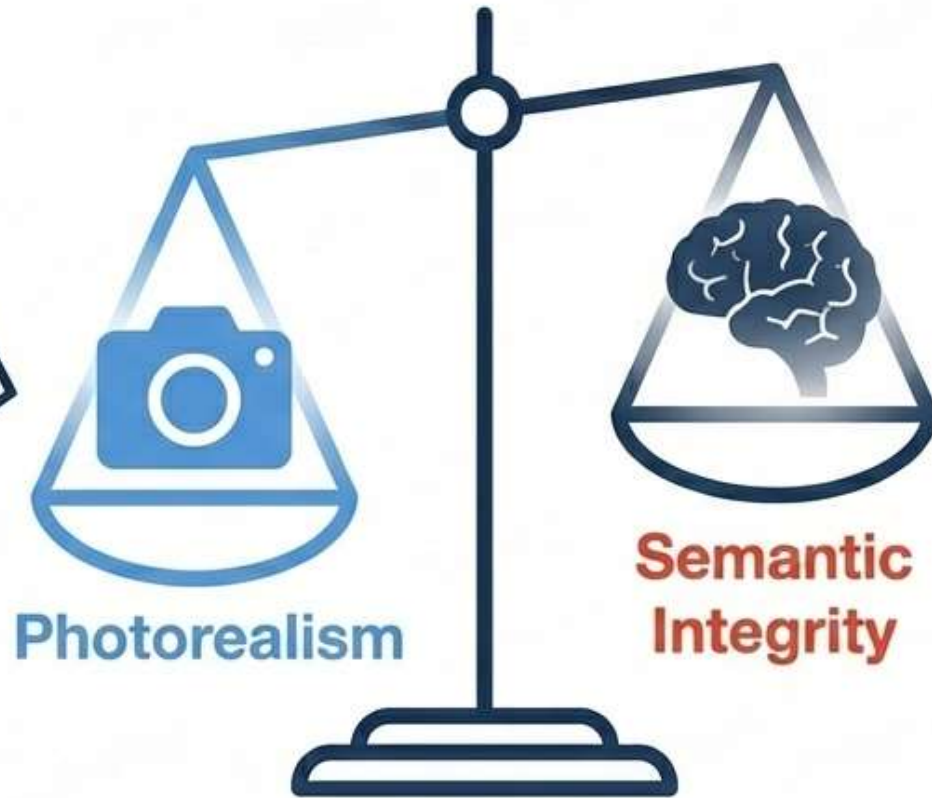
- **The Pivot:** Generative Adversarial Networks (GANs) as an infinite source of data.
- **The Promise:** Inexpensive, regulation-free, and hyper-realistic training sets [4, 5].
- **Process:** Translating raw 3D meshes (RVF10K) into real-world style (FER-2013).



The Hidden Trade-off

Photorealism vs. Semantic Integrity

- **Mathematical Non-Neutrality:**
Optimization for realism can destroy emotional micro-expressions.



- **Primary Question:**
Are we sacrificing algorithmic empathy for visual fidelity?

- **Our Hypothesis:**
Generative refinement inadvertently alters the semantic “truth” of the emotion.

Core Research Question

Beyond Simple Application

- What are the costs of “plug-and-play” GAN augmentation for FER?
- Evaluating model fairness and sensitivity using Explainable AI (XAI).
- Can we bridge the domain gap without poisoning the training well?



Three-Stage Systematic Framework

Reproducible Methodology Pipeline



Synthetic-to-Real
style translation.

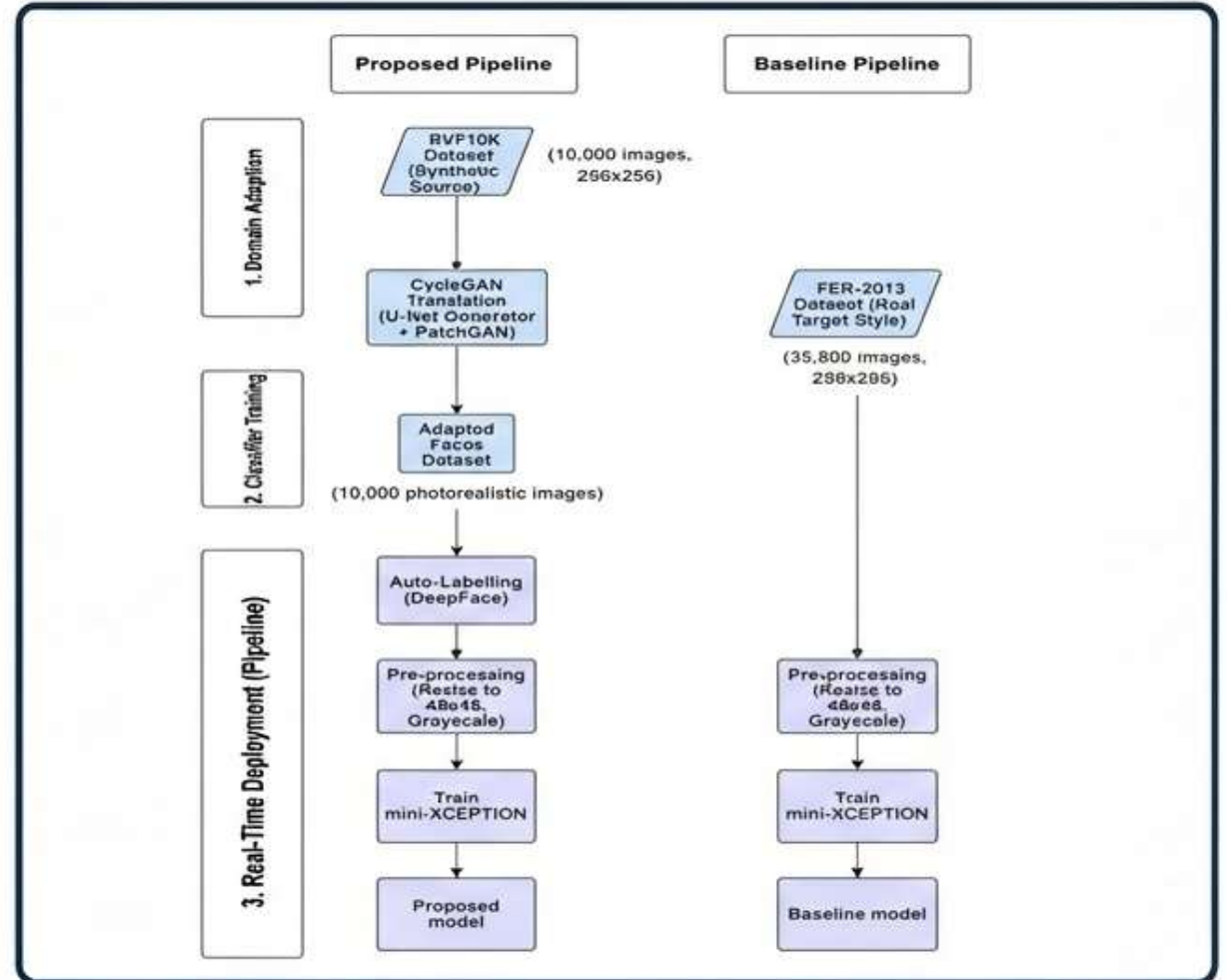
Dual mini-XCEPTION
head-to-head.

Live qualitative
inference validation.

Stage 1: Bridging the Domain Gap

CycleGAN Architecture Details

- **Source:** 10,000 synthetic images (RVF10K).
- **Generator:** U-Net for structural ROI preservation [9].
- **Discriminator:** PatchGAN for high-frequency textural realism.
- **Mapping:** Synthetic features → FER-2013 statistical distributions.



Stage 2: Dual Classifier Training

Comparative Experimental Design

- **Architecture:** Lightweight mini-XCEPTION for low latency [2].
- **Input:** 48x48 grayscale ROI extraction.
- **Baseline:** Original FER-2013 real-world data.
- **Proposed:** Auto-labeled GAN-adapted corpus.

Parameter	Baseline (FER-2013)	Proposed (GAN-Adapted)
Dataset Size	e.g., ~28k training images	e.g., ~100k+ adapted images
Labels	Manual, Noisy Real-world	Auto-labeled, GAN-refined
Augmentation Method	Standard (Crop, Flip, Rotate)	GAN-based + Standard

Stage 3: Real-Time Web Deployment

Qualitative Inference Validation

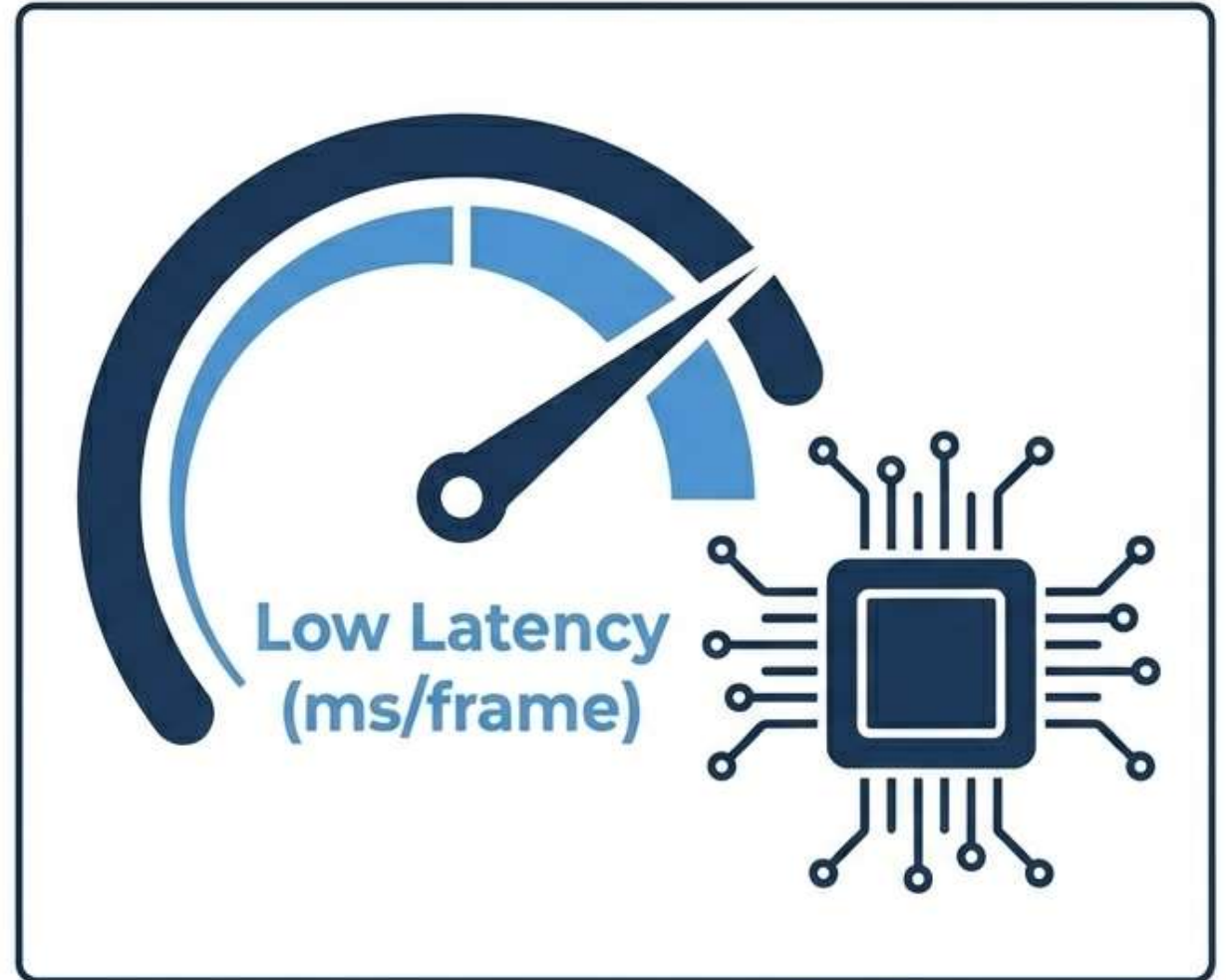


- **UI:** Streamlit progressive web application.
- **Face Detection:** Haar Cascade for rapid localization.
- **Analysis:** Frame-by-frame comparison of Baseline vs. GAN-adapted models.
- **Metrics:** Continuous confidence jitter and prediction stability.

Computational Efficiency for HCI

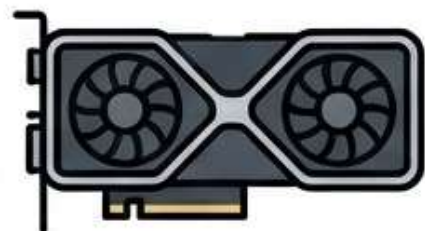
Designed for Edge Responsiveness

- **Priority:** Low latency for interactive empathy.
- **Lightweight Models:** Fast inference without massive compute overhead.
- **Scalability:** Optimized for diverse hardware environments.



Reproducibility & Hardware

Rigorous Training Constraints



Hardware:
NVIDIA RTX
3080 GPU.



Stack:
TensorFlow 2.10,
Python 3.8.



Domain
Adaptation:
18 Hours

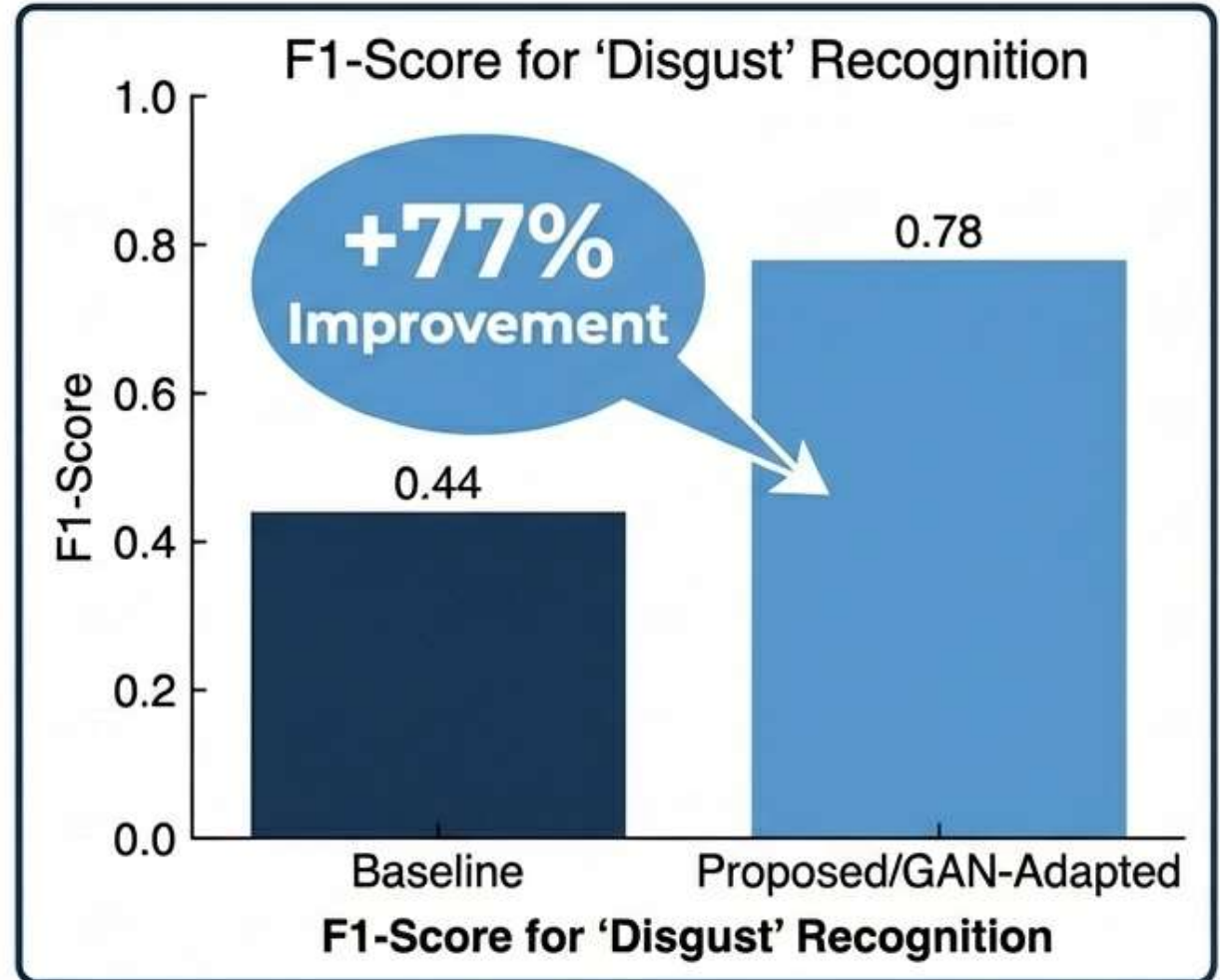


Classifier
Training:
< 2 Hours

The Success Story: Minority Rescue

Dramatic Gains in 'Disgust' Recognition

- **The Win:** F1-Score for rare 'Disgust' class jumped from **0.44 to 0.78**.
- **True Positive Rate:** Soared from **30% to 68%**.
- **Conclusion:** GAN-based augmentation effectively rescues imbalanced classes [6].



The Statistical Illusion

Dark Side of Macro-Averages

- **Overall Gain:** F1-score improved marginally from 0.64 to 0.66.
- **The Risk:** Top-level metrics **mask catastrophic subclass failures.**



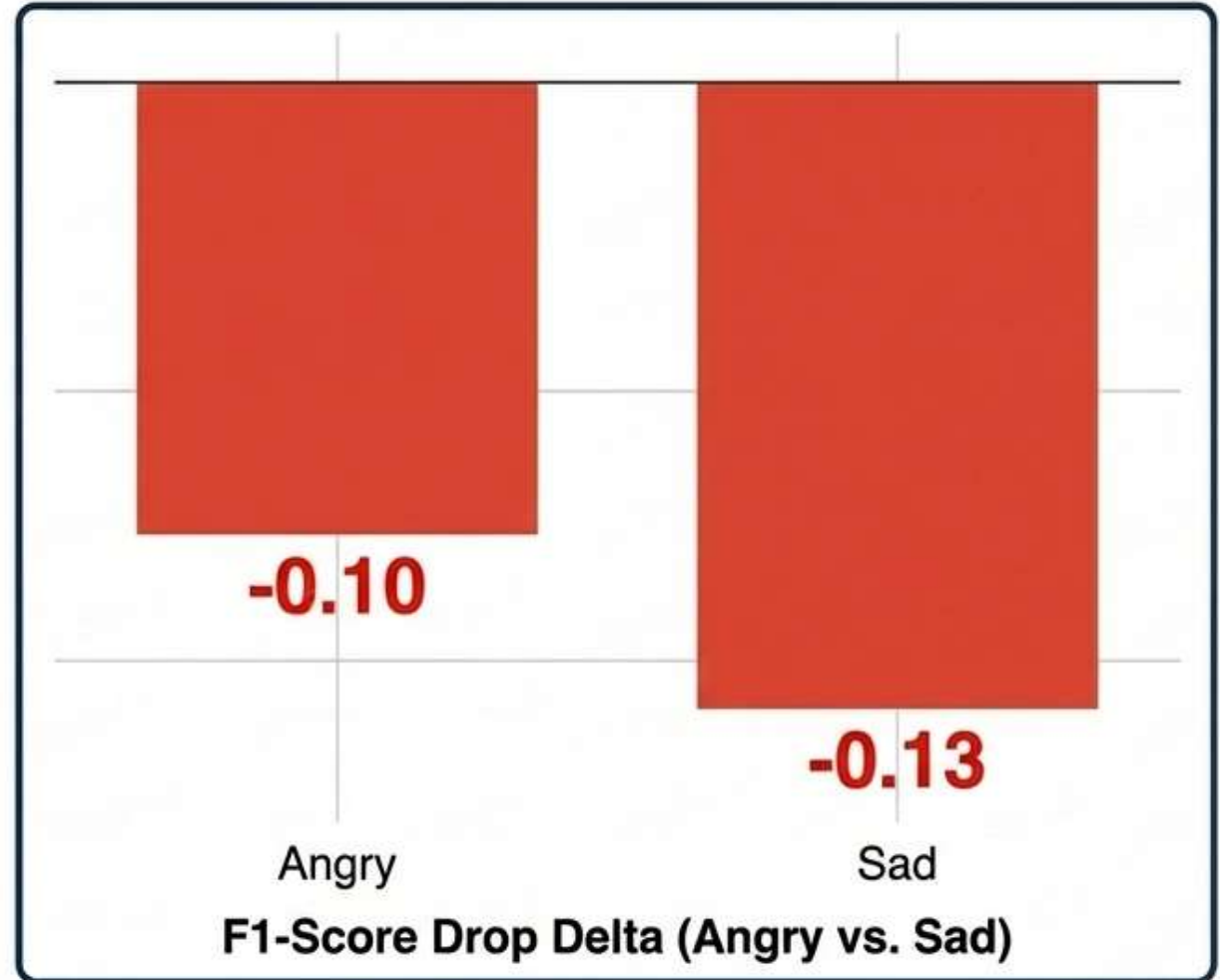
Top-level metrics mask catastrophic subclass failures.

- **Lesson:** Aggregates hide that the 'semantic cost' of domain adaptation.

The Critical Performance Paradox

Destruction of Nuance

- **The Trade-off:** Gains in Disgust cost sensitivity in Anger and Sadness.
- **Angry:** F1 dropped from **0.63** to **0.53**.
- **Sad:** F1 dropped from **0.60** to **0.47**.
- **Cause:** GAN “smoothing” filters out low-intensity negative cues.



The Smoking Gun

Confusion Matrix Analysis

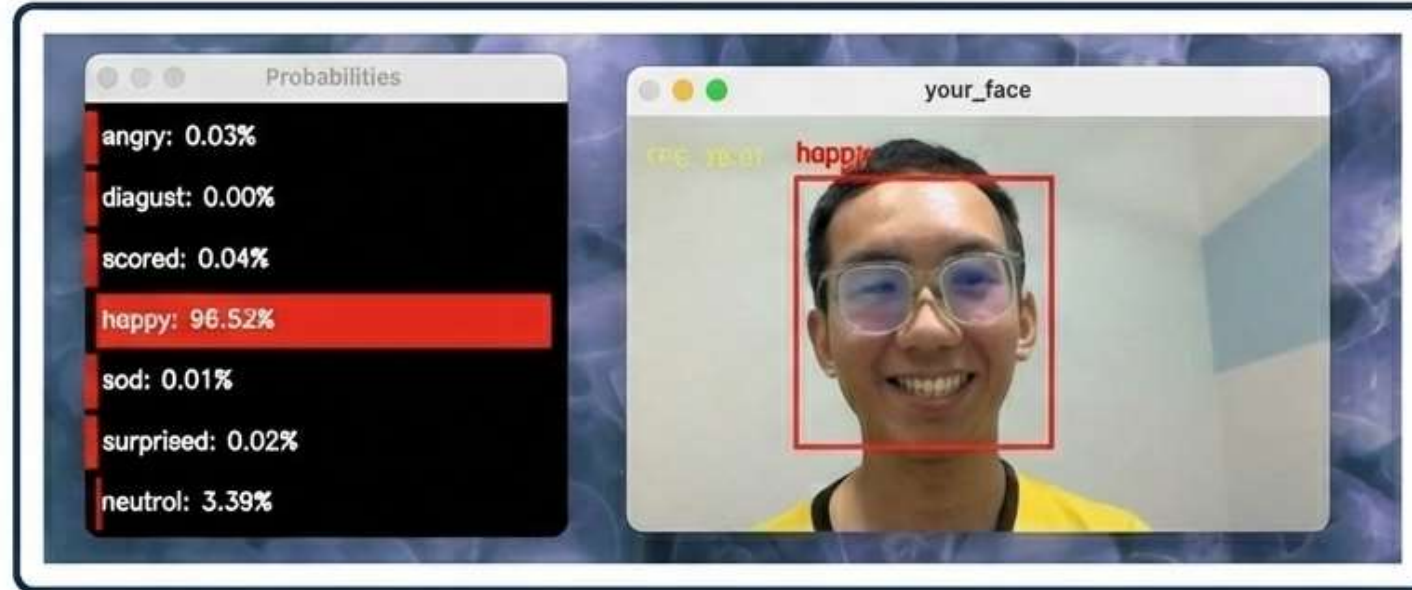
- **Angry** misclassified as Neutral: 42%.
- **Sad** misclassified as Neutral: 44%.
- **Insight:** GAN mimics the majority class bias (Neutral) of synthetic sets [7].

Table 3. Confusion Matrix for Proposed GAN-Adapted Model (%)

	Neutral	Happy	Sad	Angry
Neutral	92%	3%	2%	3%
Happy	5%	94%	1%	0%
Sad	44%	2%	51%	3%
Angry	42%	1%	4%	53%

Live Inference: The 'Happy' Bias

High Confidence, Low Sensitivity



Success

- High intensity (Happy) classified **with >96% confidence.**

Failure

- Total blindness to subtle face cues.

Behavior

- Model defaults to 'Happy' or 'Neutral' on ambiguous input.

Visualizing Semantic Loss

The Drastic XAI Perspective

Erosion of Micro-expressions

Original

GAN Generated

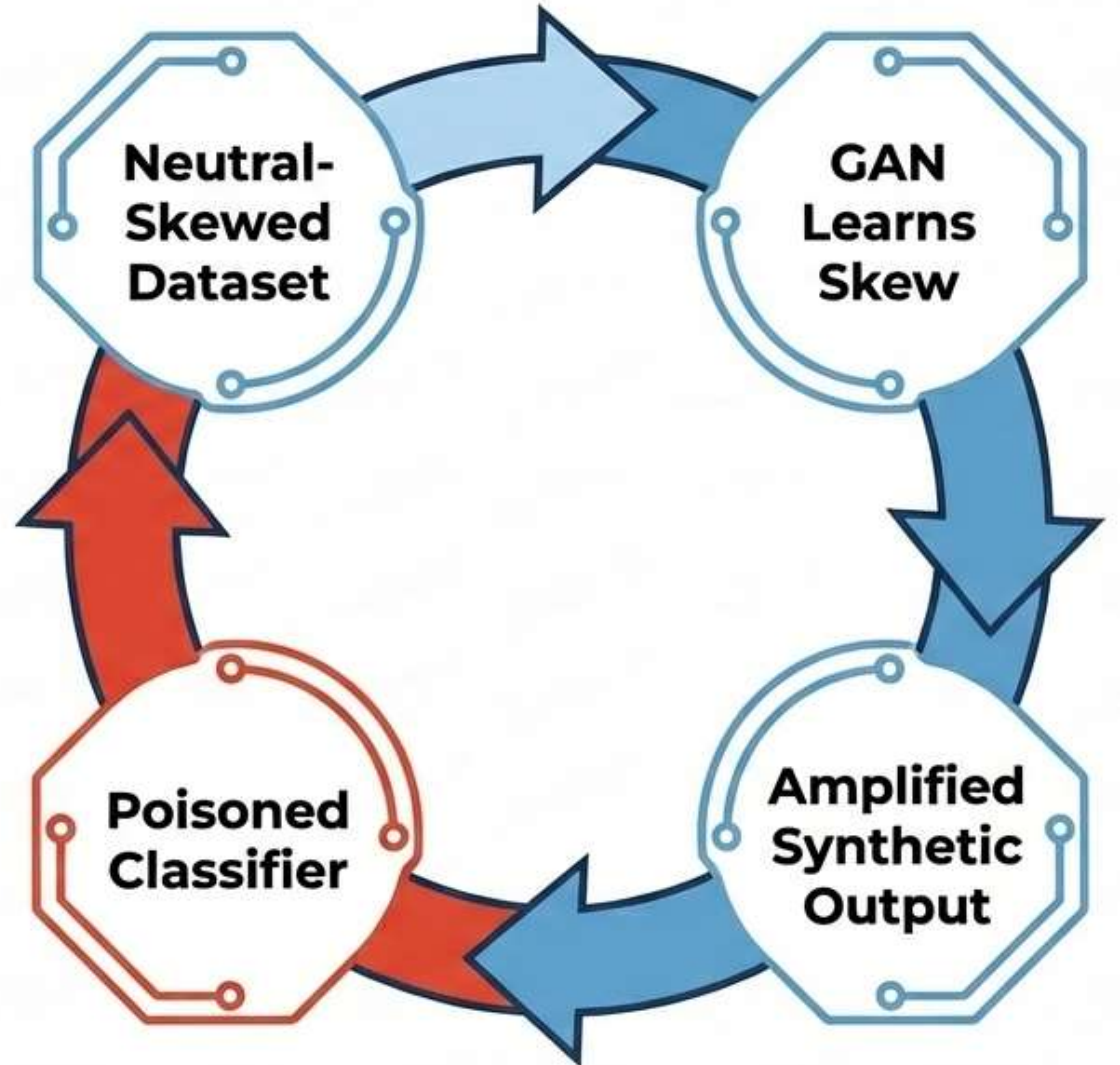


- **Optimization:** Pursuit of “Realism” acts as a destructive filter.
- **Detail Erosion:** GANs “smooth” micro-tensions in eyes and brow.
- **Result:** Machine-readable signals are lost to human-readable aesthetics.

Algorithmic Bias Amplification

The Feedback Loop

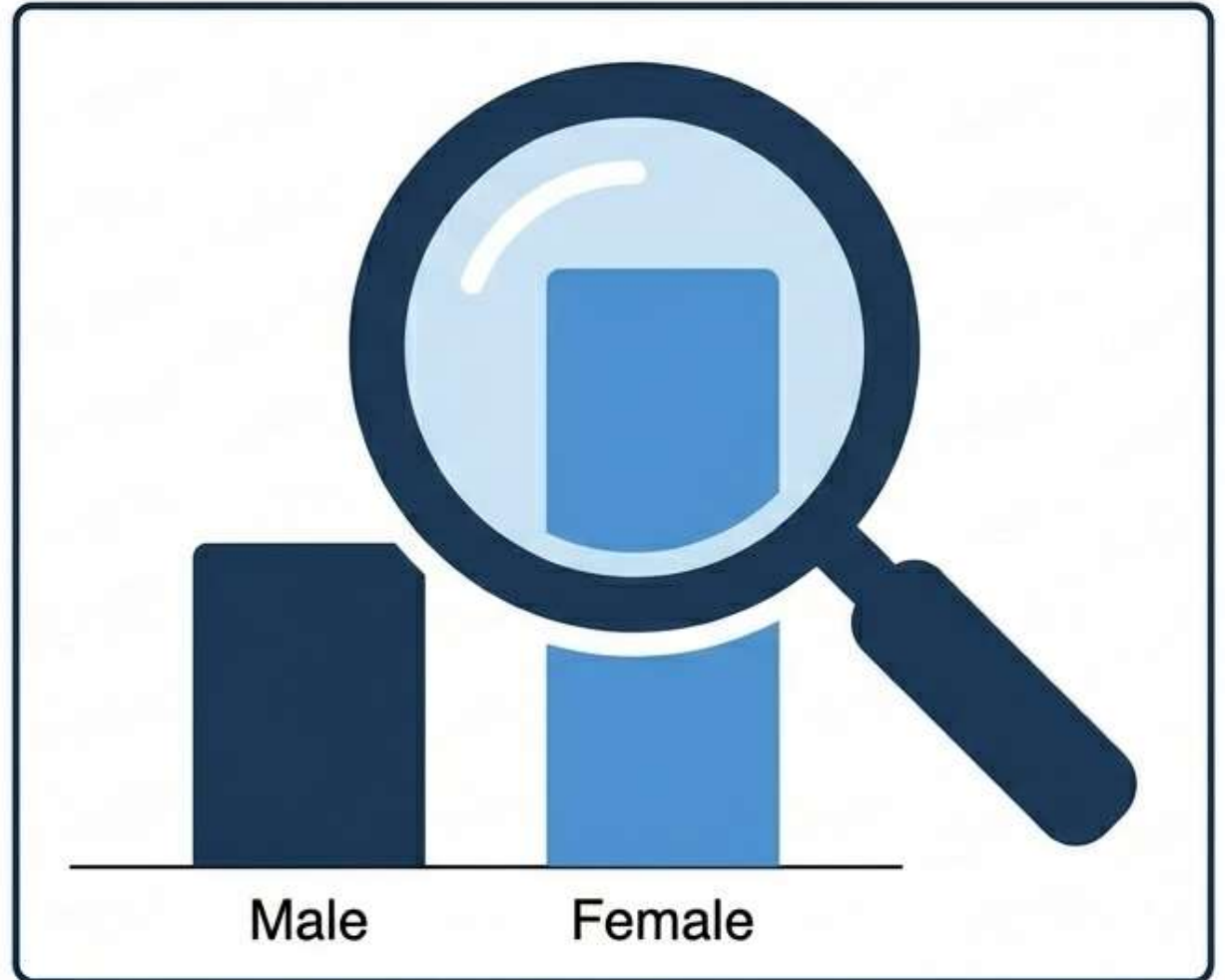
- **Dataset Bias:** FER-2013 is heavily 'Neutral'-skewed.
- **GAN Cycle:** Generative model learns to mimic this skew.
- **Amplification:** Output data is even more skewed than the input.



Beyond Averages

Demographic Subgroup Disparities

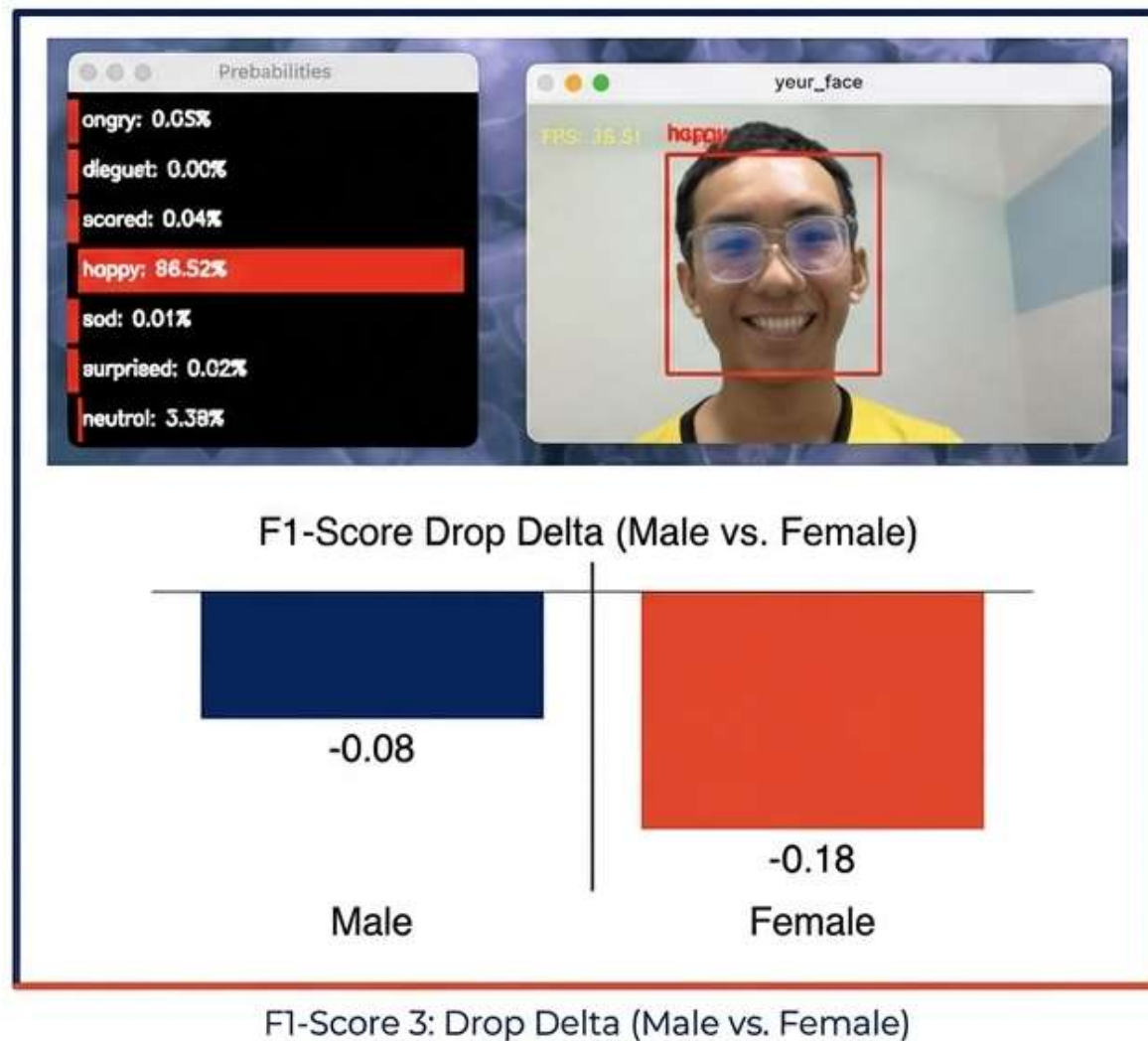
- **Question:** Who bears the cost of this bias amplification?
- **Analysis:** Splitting performance metrics by gender.
- **Finding:** Bias is not distributed evenly across populations.



The Gender Disparity Penalty

Bias Amplification in Action

- Angry F1 Drop (Male): **-0.08**
- Angry F1 Drop (Female): **-0.18**
- Disparity: Performance drop was **2.25x worse** for female subjects.



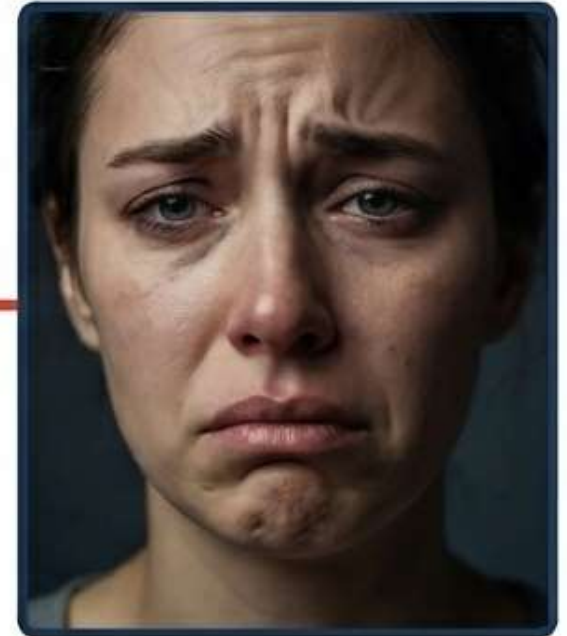
Case Study: AI Telehealth

Real-World Risks of Misclassification

- **Scenario:** Remote Patient Monitoring for Mental Health.
- **Reality:** User exhibits subtle “Sad” distress.
- **AI Outcome:** System logs user as “Neutral/Fine” due to GAN smoothing.
- **Risk:** Critical triage failure in life-saving systems.




Error



**Risk: Critical triage failure
in life-saving systems.**

Emotionally Blind by Design

A Failure of Fairness in HCI

SYSTEMIC EMOTIONAL BLINDNESS

- **Consequence:** Systems become systematically less sensitive to specific demographics.
- **Deployment Risk:** Trust erosion and ethical violations in predictive analytics.
- **Verdict:** Unsupervised generation can lead to accidental emotional blindness.

The Accessibility Crisis

Marginalized Communities Impact



- **Reliance:** Deaf/Hard-of-Hearing rely heavily on visual sentiment cues [23].
- **Severed Bridge:** Biased generation disrupts essential AR/XR communication.
- **HCI Goal:** Creating equitable tools, not reflecting systemic imperfections.

The Illusion of “More Data”

A Benchmark Cautionary Tale

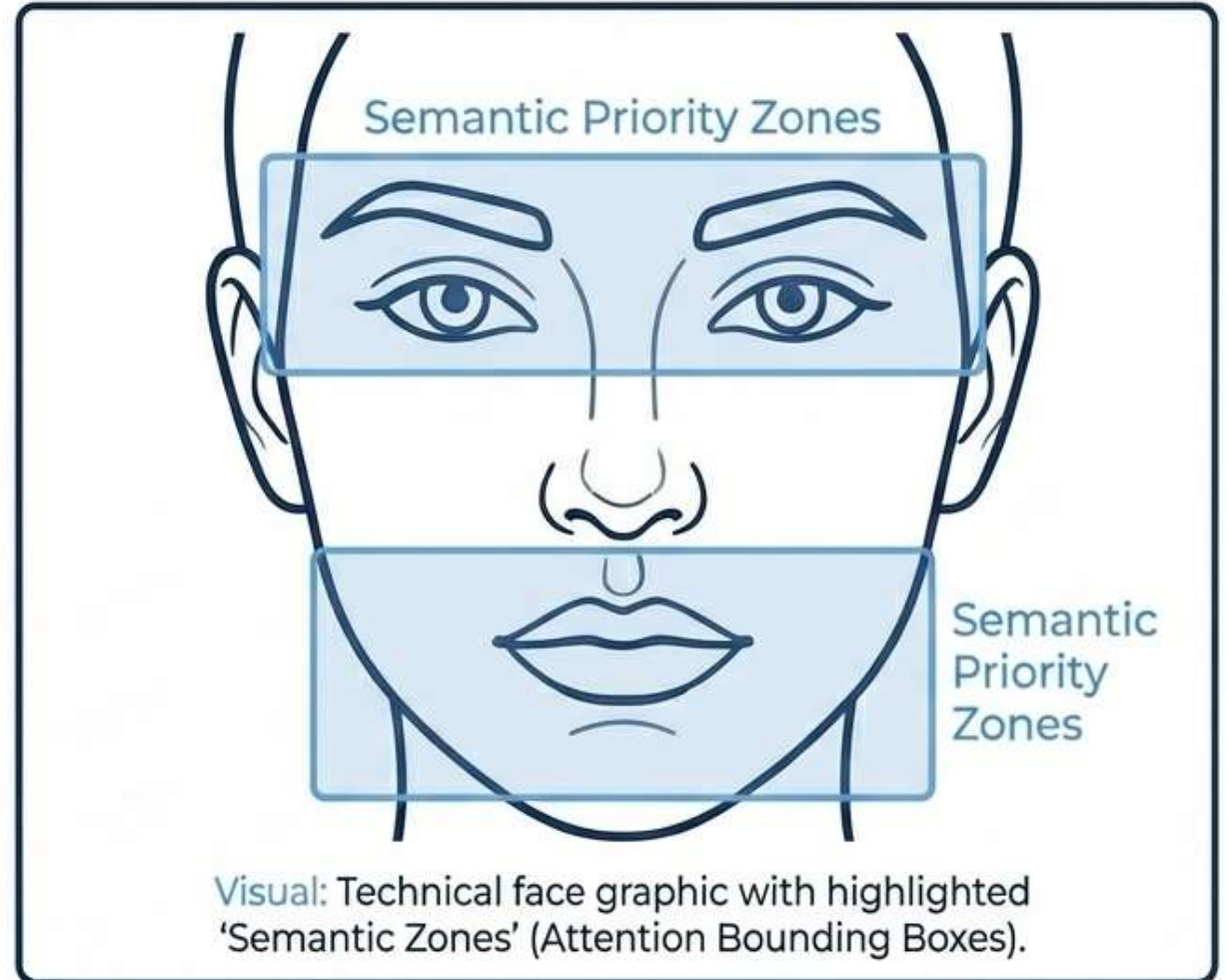
- **Takeaway:** “More data” $\neq$ “Better models.”
- **Peril:** Plug-and-play generation can poison your training pipeline.
- **Warning:** Traditional GANs are not a neutral pre-processing step.



Solution 1: Spatial Attention

Preserving Semantic Cues

- **Mechanism:** Implement Attention-guided GANs [9].
- **Constraint:** Force the generator to prioritize eyes, mouth, and eyebrows.
- **Benefit:** Protects anatomical structural integrity during translation.



Solution 2: Fairness-Aware Loss

Penalizing Bias Transfer

- **Innovation:** Integrate **Demographic Parity** into the Loss Function [10].
- **Objective:** Explicitly penalize the GAN for amplifying subgroup bias.
- **Outcome:** Equitable performance across all user demographics.

$$L_{total} = L_{GAN} + \lambda L_{Fairness}$$

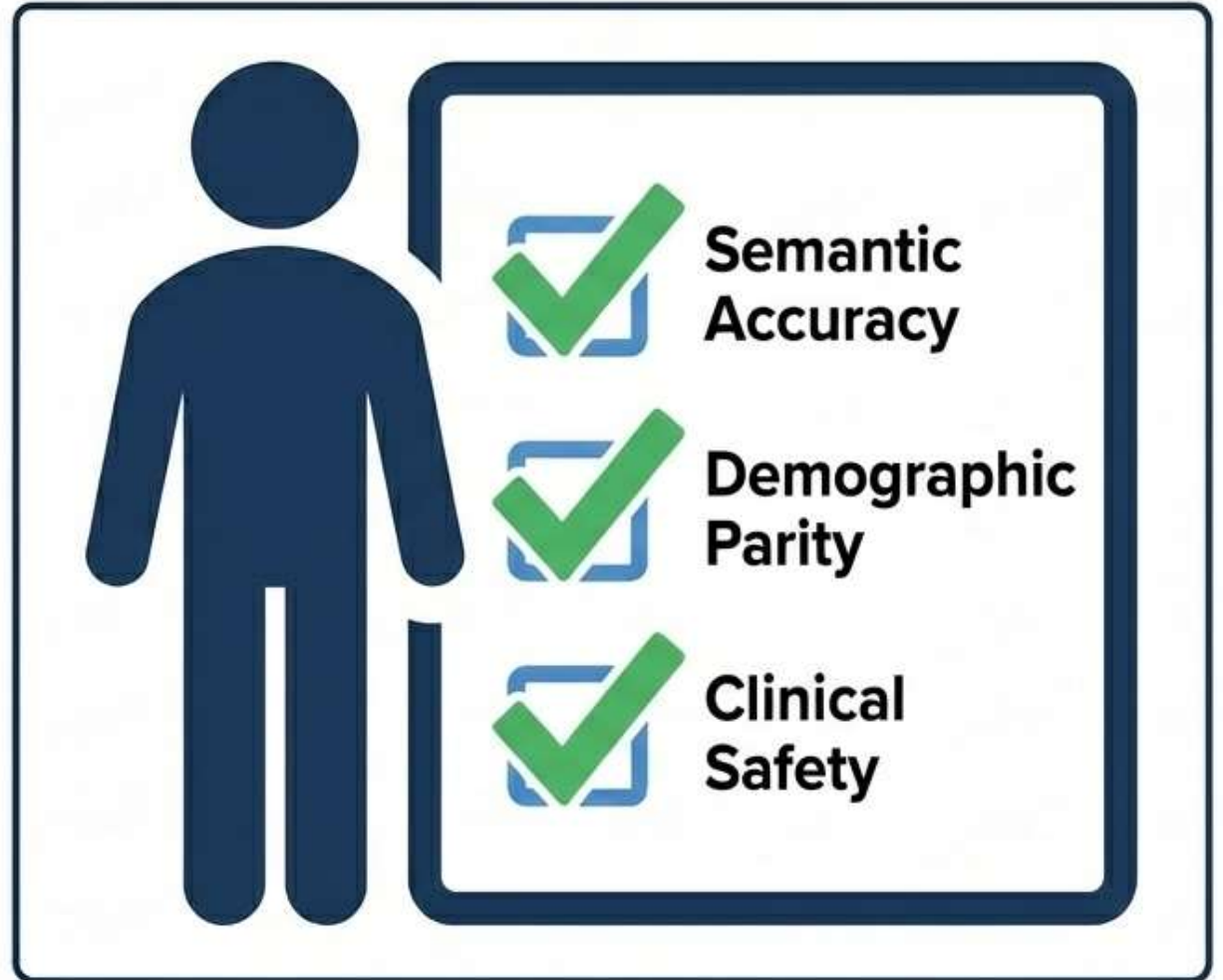
[10]



Solution 3: Human-in-the-Loop

Rigorous Curation & Validation

- **Shift:** From infinite generation to critical curation.
- **Auditing:** Expert validation of semantic truth in synthetic samples.
- **Security:** Preventing systemic vulnerabilities in health/safety AI.



Commercial Risks of Biased AI

The Cost of Emotionally Blind Tech

- **Regulations:** Synthetic data is not a shield against EU AI Act fines.
- **Brand Hazard:** Tone-deaf AI outputs lead to immediate user churn.
- **Market Loss:** Failure to serve 50% of the population (gender bias).



The Market for Empathetic AI

Growth Through Equity

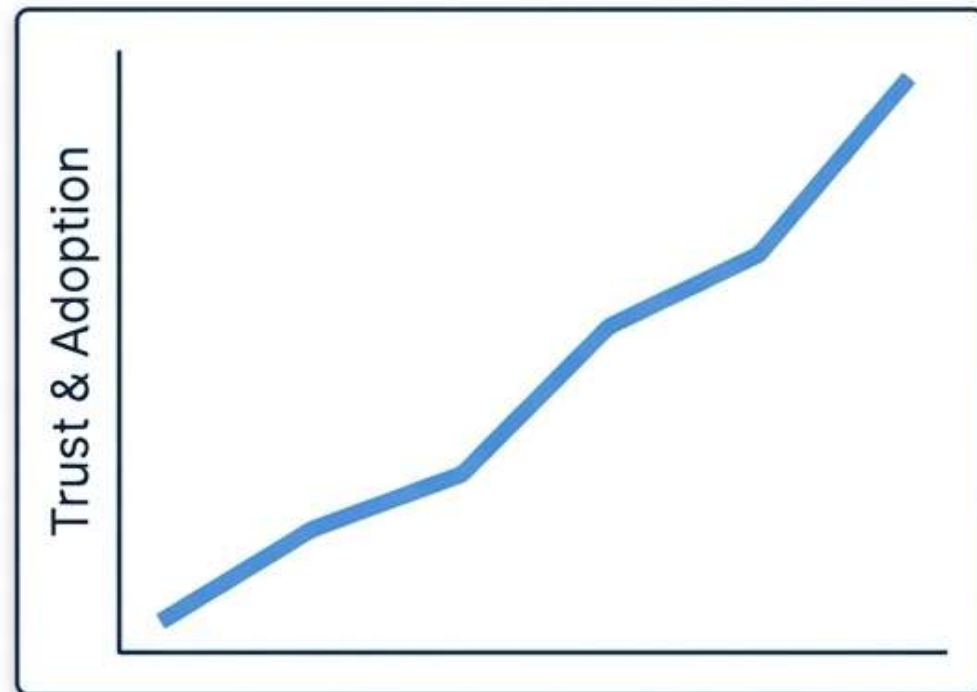
- Market Potential: Affective Computing exceeding **\$200 Billion** by 2030.
- Key Sectors: Telehealth, EdTech, Automotive UX.
- **Differentiator:** Trust and fairness as core commercial features.



Conclusion & The Fairness Dividend

Enhancing Human Interaction

- **Finding:** GAN adaptation is a double-edged sword: rescue one class, fail others.
- **Vision:** Shift from "data quantity" to "bias-mitigated quality."
- **Final Note:** Let AI enhance interaction, not reflect our imperfections.



Q&A | Kevin Lee | Faculty of IT, TARUMT